FODS ASSIGNMENT 1

Name – **Atharva Shivaji Dabkekar**
Roll no. – **2090139**
PRN - **1251090357**
Division – **S.Y A**
Batch – **2**
DATE – **14/07/2026** TIME – **8:00 TO 10:00**

QUESTION) Practical 1: Algorithm Design and Complexity Analysis

Design algorithms using pseudocode and flowcharts for real-world problems. Analyse their time and space complexities using Big-O, Theta, and Omega notations.

Write a C/C++/Python program to search a given element in an array using Linear Search. Display the position if found; otherwise display "Element not found". Analyse the algorithm using Big-O, Theta, and Omega notations

ANSWER)

Linear search is simple search technique used to find a element it search one element at a time until the element is found or it reaches the end .

Algorithm:

Step 1] Start

Step 2] Read the size of the array n.

Step 3] Read the array elements.

Step 4] Read the element to be searched (key).

Step 5] Set i = 0.

Step 6] Repeat while i < n

If A[i] == key

Display "Element found at position i + 1"

Stop

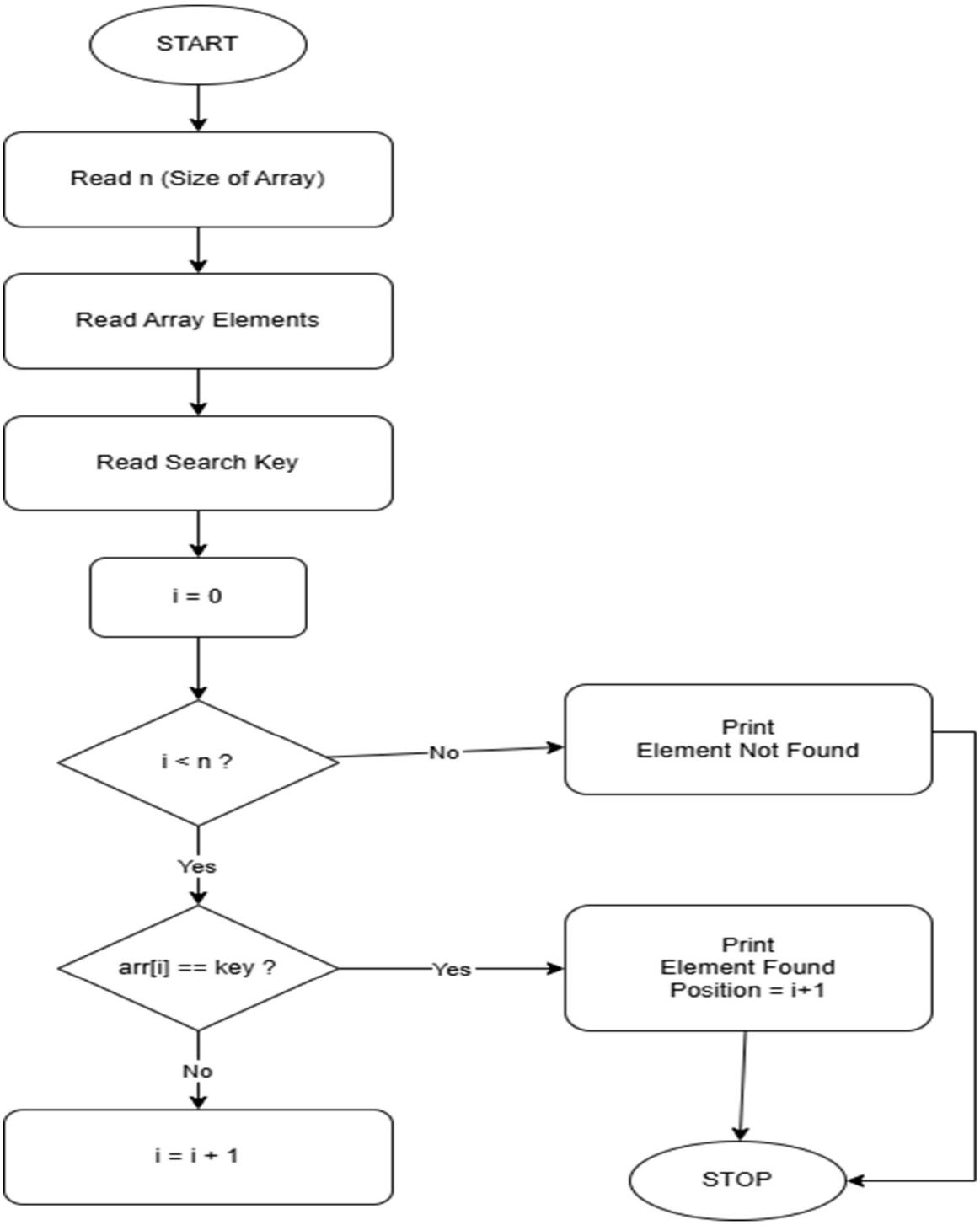
Else

Increment i

Step 7]Display "Element not found"

Step 8]Stop

Flowchart:



Code:

```
#include <stdio.h>
int main()
{
   int n, i, key;

   printf("Enter size of array: ");
   scanf("%d", &n);

   int arr[n];

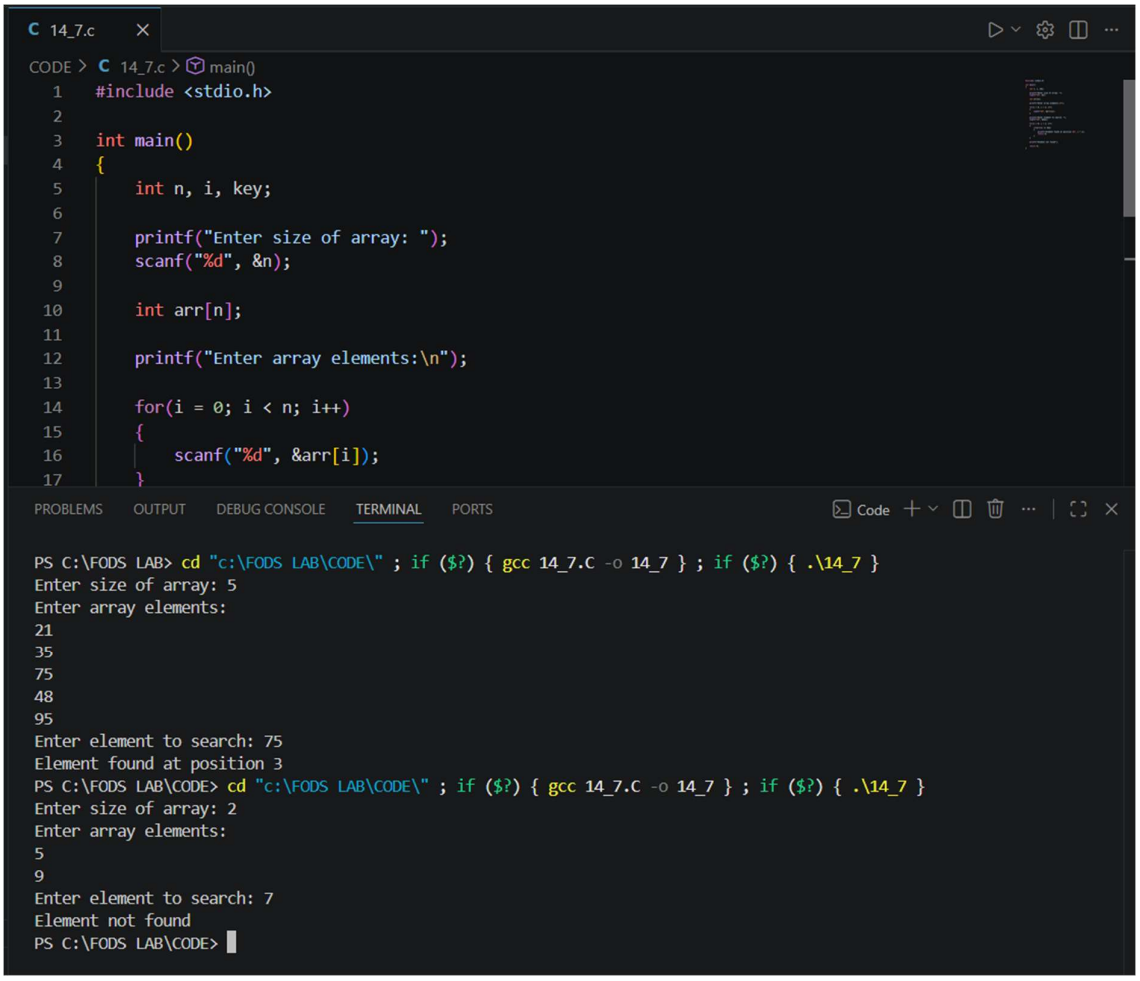
   printf("Enter array elements:\n");

   for(i = 0; i < n; i++)
   {
      scanf("%d", &arr[i]);
   }

   printf("Enter element to search: ");
   scanf("%d", &key);

   for(i = 0; i < n; i++)
   {
      if(arr[i] == key)
      {
         printf("Element found at position %d", i + 1);
         return 0;
      }
   }
printf("Element not found");
 return 0;
}
```

Output:



```
C 14_7.c
CODE > C 14_7.c > main()
#include <stdio.h>

int main()
{
    int n, i, key;

    printf("Enter size of array: ");
    scanf("%d", &n);

    int arr[n];

    printf("Enter array elements:\n");

    for(i = 0; i < n; i++)
    {
        scanf("%d", &arr[i]);
    }

PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS

PS C:\FODS LAB> cd "c:\FODS LAB\CODE\" ; if ($?) { gcc 14_7.C -o 14_7 } ; if ($?) { .\14_7 }
Enter size of array: 5
Enter array elements:
21
35
75
48
95
Enter element to search: 75
Element found at position 3
PS C:\FODS LAB\CODE> cd "c:\FODS LAB\CODE\" ; if ($?) { gcc 14_7.C -o 14_7 } ; if ($?) { .\14_7 }
Enter size of array: 2
Enter array elements:
5
9
Enter element to search: 7
Element not found
PS C:\FODS LAB\CODE>
```

Time complexity And Space complexity:

CASE	NOTATION	COMPLEXITY
Best case	Ω	Ω(1)
Average case	θ	θ (n)
Worst case	O	O(n)